Extra Bite #10: Diodes (Zener, Schottky, PIN, Varactor, LED)

Specialty diodes -- what each one is for, and why a junction diode dies from too much current.

Group E6B · 11 pool questions · Standalone · 5-10 minute presentation · Practice exams at eham.net/exams

Why You Care

A plain rectifier diode is just a one-way valve, but the diode family is wide. Each specialty diode trades one property of an ordinary PN junction for a useful behavior: voltage reference, voltage-controlled capacitance, RF switching, microwave detection, light. Extra asks you to recognize each one by description and one or two distinctive properties.

The Diode Zoo

Type	Most useful characteristic	Common use
Zener	**Constant voltage drop under varying current.** (E6B01) Engineered to break down at a precise reverse voltage and hold there.	Voltage reference; protection clamp.
Schottky	**Lower forward voltage drop** than a silicon junction diode (E6B02). Built as a **metal-semiconductor junction** (E6B08), very fast switching, no carrier-storage delay.	Power-supply rectifier; **VHF/UHF mixer or detector** (E6B06).
PIN	**Low junction capacitance** makes it useful as an RF switch (E6B05). Attenuation is controlled by **forward DC bias current** (E6B11).	RF switch, T/R switching, RF attenuator.
Varactor	Voltage-controlled capacitance. Reverse-biased PN junction whose depletion-region width (and so its capacitance) changes with applied voltage. (E6B04)	Voltage-tuned oscillators; AFC.
LED	Forward voltage drop is set by the semiconductor's **band gap** (E6B03). Bigger band gap = higher forward V = bluer light.	Indicators; emitter side of optoisolators.
Point-contact	Very small junction area, very low capacitance. Standard early **RF detector** (E6B09). Still seen in some microwave detectors.	RF detector.

One Failure Mode

What kills a junction diode at excessive forward current? Excessive junction temperature. (E6B07) Current dumps power into the tiny junction; the junction overheats; the semiconductor structure degrades and the diode goes open or short.

This is why power rectifiers come with current ratings, and why a heatsink is often the difference between "working" and "smoke."

Figure E6-2 (Schottky Symbol)

The only diode-symbol question E6B asks is **which is the Schottky in Figure E6-2**. The Schottky symbol has the diode triangle with a small "S"-shaped (or hook-shaped) bar at the cathode instead of a plain straight line.

Figure E6-2

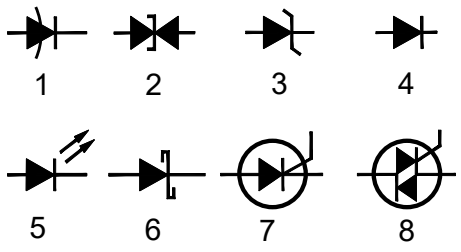


Figure E6-2 shows six numbered diode symbols. Memorize:

- **Symbol 6 = Schottky diode.** (E6B10) The cathode bar has the distinctive "S" hook on it.

Other symbols in the figure are a plain rectifier diode, Zener, varactor, tunnel diode, LED. The 2024-2028 pool only tests *which one is the Schottky*, so anchor on the hook-shaped cathode line.

Mental Cheat Sheet

- Zener → constant **voltage** under varying current. (Reference.)
- Schottky → metal-semiconductor junction, **low forward drop**, **fast**. Common: power supply rectifier; VHF/UHF mixer/ detector.
- PIN → low junction capacitance → RF switch. Controlled by **forward DC bias current**.
- Varactor → voltage-controlled **capacitance**.
- LED forward V is set by the **band gap**.

Common Traps

- Zener = constant **voltage**, not constant current.
- Schottky's advantage over silicon as a rectifier is **low forward drop**, not "higher reverse breakdown."
- PIN attenuation is controlled by **DC bias current**, not by RF voltage.
- Varactor is a **voltage-controlled capacitor**, not a current source.
- A junction diode dies from **temperature**, not from reverse voltage alone (that is a different failure mode).

- Point-contact diode → **RF detector**.
- Excessive forward current kills a junction diode by **excessive junction temperature**.
- Figure E6-2: Schottky = symbol **6** (hook on the cathode bar).

- Point-contact diodes are **RF detectors**, not constant-voltage sources.

Takeaway: The specialty diodes each give up one ordinary-diode property for a useful behavior: Zener for voltage reference, Schottky for low drop and speed, PIN for RF switching, varactor for tunable capacitance, LED for light, point-contact for high-frequency detection. The one failure-mode question to remember: too much forward current means too much heat at the junction.

Bite 10 Question Bank (E6B)

E6B01. What is the most useful characteristic of a Zener diode?
A. A constant current drop under conditions of varying voltage.
B. A constant voltage drop under conditions of varying current.
C. A negative resistance region.
D. An internal capacitance that varies with the applied voltage.

E6B02. Which characteristic of a Schottky diode makes it a better choice than a silicon junction diode for use as a power supply rectifier?
A. Much higher reverse voltage breakdown.
B. More constant reverse avalanche voltage.
C. Longer carrier retention time.
D. Lower forward voltage drop.

E6B03. What property of an LED's semiconductor material determines its forward voltage drop?
A. Intrinsic resistance.
B. Band gap.
C. Junction capacitance.
D. Junction depth.

E6B04. What type of semiconductor device is designed for use as a voltage-controlled capacitor?
A. Varactor diode.
B. Tunnel diode.
C. Silicon-controlled rectifier.
D. Zener diode.

E6B05. What characteristic of a PIN diode makes it useful as an RF switch?
A. Extremely high reverse breakdown voltage.
B. Ability to dissipate large amounts of power.
C. Reverse bias controls its forward voltage drop.
D. Low junction capacitance.

E6B06. Which of the following is a common use of a Schottky diode?
A. In oscillator circuits as the negative resistance element.
B. As a variable capacitance in an automatic frequency control circuit.
C. In power supplies as a constant voltage reference.
D. As a VHF/UHF mixer or detector.

E6B07. What causes a junction diode to fail from excessive current?
A. Excessive inverse voltage.
B. Excessive junction temperature.
C. Insufficient forward voltage.
D. Charge carrier depletion.

E6B08. Which of the following is a Schottky barrier diode?
A. Metal-semiconductor junction.
B. Electrolytic rectifier.
C. PIN junction.
D. Thermionic emission diode.

E6B09. What is a common use for point-contact diodes?
A. As a constant current source.
B. As a constant voltage source.
C. As an RF detector.
D. As a high-voltage rectifier.

E6B10. In Figure E6-2, which is the schematic symbol for a Schottky diode?
A. 1.
B. 6.
C. 2.
D. 3.

E6B11. What is used to control the attenuation of RF signals by a PIN diode?
A. Forward DC bias current.
B. A variable RF reference voltage.
C. Reverse voltage larger than the RF signal.
D. Capacitance of an RF coupling capacitor.

Answer Key

E6B01: B	E6B02: D	E6B03: B	E6B04: A
E6B05: D	E6B06: D	E6B07: B	E6B08: A
E6B09: C	E6B10: B	E6B11: A	